TikuOS Bus Interfaces: I2C & SPI

Simple. Ubiquitous. Intelligence, Everywhere.

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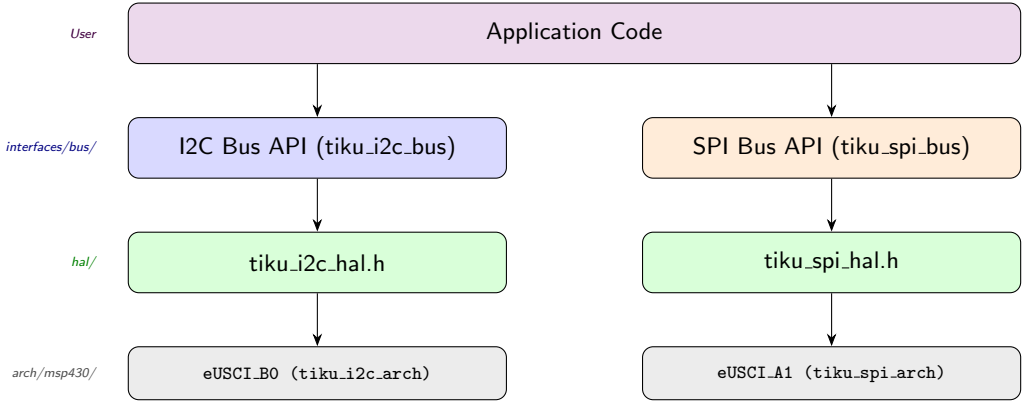
Outline

1. Architecture Overview
2. I2C Bus Interface
   - I2C Protocol
   - I2C API
   - I2C Architecture
3. SPI Bus Interface
   - SPI Protocol
   - SPI API
   - SPI Architecture
4. Usage Examples
5. File Organization
6. Summary

Three-Layer Bus Abstraction



Design Principle

Platform-independent API in `interfaces/bus/`. Board-specific pin routing and clock prescalers in board headers. One arch implementation works across all MSP430FR variants.

I2C vs SPI at a Glance

	I2C	**SPI**
Wires	2 (SDA + SCL)	3 + 1/device (SCLK + SIMO + SOMI + CS)
Addressing	7-bit slave address on bus	Chip select GPIO per device
Duplex	Half-duplex	Full-duplex
Speed	100 kHz / 400 kHz	Up to SMCLK/2 (4 MHz @ 8 MHz)
MSP430 Module	eUSCI_B0	eUSCI_A1
Typical Use	Sensors, EEPROMs, RTCs	Flash, displays, radio transceivers

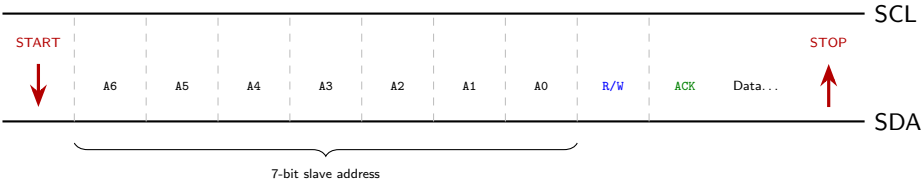
FR5969 LaunchPad I2C Pins

P1.6 = SDA, P1.7 = SCL
(eUSCI_B0, SEL1=1 SEL0=0)

FR5969 LaunchPad SPI Pins

P2.5 = CLK, P2.6 = SIMO, P2.7 = SOMI
(eUSCI_A1, SEL1=1 SEL0=0)

I2C Protocol Basics



Transaction types:

- **Write**: START → addr+W → data → STOP
- **Read**: START → addr+R → data → NACK → STOP
- **Write-Read**: write → repeated START → read (register-read pattern)

Key characteristics:

- Multi-master, multi-slave on same bus
- ACK/NACK feedback per byte
- Clock stretching by slave
- Open-drain lines with pull-ups

I2C Configuration & Status Codes

interfaces/bus/tiku_i2c_bus.h

```
/* Speed modes */
#define TIKU_I2C_SPEED_STANDARD 0
#define TIKU_I2C_SPEED_FAST     1

/* Status codes */
#define TIKU_I2C_OK           0
#define TIKU_I2C_ERR_NACK    (-1)
#define TIKU_I2C_ERR_TIMEOUT (-2)
#define TIKU_I2C_ERR_BUSY    (-3)
#define TIKU_I2C_ERR_PARAM   (-4)

/* Configuration */
typedef struct tiku_i2c_config {
    uint8_t speed;
} tiku_i2c_config_t;
```

Speed modes:

- STANDARD: 100 kHz (prescaler 80)
- FAST: 400 kHz (prescaler 20)
- Prescalers from 8 MHz SMCLK

Error Handling

Every bus operation returns a status code.
NACK: slave didn't respond — wrong address or busy.
TIMEOUT: bus stuck — hardware issue.
All errors auto-issue STOP to release bus.

I2C API Functions

Function	Description
`tiku_i2c_init(config)`	Initialize eUSCI_B0 in I2C master mode
`tiku_i2c_close()`	Place peripheral in reset
`tiku_i2c_write(addr, buf, len)`	START → addr+W → data → STOP
`tiku_i2c_read(addr, buf, len)`	START → addr+R → data → NACK → STOP
`tiku_i2c_write_read(addr, tx, tx_len, rx, rx_len)`	Write then repeated-START read (register access)

Typical register-read pattern

```
/* Read 2 bytes from register 0x05 of device at address 0x48 */
uint8_t reg = 0x05;
uint8_t buf[2];
int rc = tiku_i2c_write_read(0x48, &reg, 1, buf, 2);
if (rc != TIKU_I2C_OK) {
    /* handle error */
}
```

I2C Board & Device Macros

devices/tiku_device_fr5969.h

```
/* eUSCI peripheral availability */
#define TIKU_DEVICE_HAS_EUSCIB0 1
#define TIKU_DEVICE_HAS_EUSCIB1 1
```

boards/tiku_board_fr5969_launchpad.h

```
/* I2C on eUSCI_B0:
   P1.6 = SDA, P1.7 = SCL */
#define TIKU_BOARD_I2C_PINS_INIT() \
    do { P1SEL1 |= BIT6 | BIT7; \
         P1SEL0 &= ~(BIT6 | BIT7);\
    } while(0)

/* Prescalers from 8 MHz SMCLK */
#define TIKU_BOARD_I2C_BRW_100K 80
#define TIKU_BOARD_I2C_BRW_400K 20
```

Adding a new board:

1. Create board header in `boards/`
2. Define `TIKU_BOARD_I2C_PINS_INIT()` with correct port and pin SEL bits
3. Define prescaler macros for the board's SMCLK frequency
4. Add `#elif` in `tiku_device_select.h`

Pin Function Select

MSP430FR5969 uses SEL1:SEL0 = 1:0 for eUSCI_B0 alternate function on P1.6/P1.7.

I2C Arch: Initialization

arch/msp430/tiku_i2c_arch.c — tiku_i2c_arch_init()

```
int tiku_i2c_arch_init(const tiku_i2c_config_t *config)
{
    /* Select I2C function on board-specific SDA/SCL pins */
    TIKU_BOARD_I2C_PINS_INIT();

    /* Put eUSCI_B0 in reset before configuration */
    UCB0CTLW0 = UCSWRST;

    /* I2C master mode, synchronous, SMCLK source */
    UCB0CTLW0 |= UCMODE_3 | UCMST | UCSSEL__SMCLK | UCSYNC;

    /* Set clock prescaler for requested speed */
    if (config->speed == TIKU_I2C_SPEED_FAST) {
        UCB0BRW = TIKU_BOARD_I2C_BRW_400K;
    } else {
        UCB0BRW = TIKU_BOARD_I2C_BRW_100K;
    }

    UCB0CTLW1 = UCASTP_0;        /* Manual STOP (no auto-stop) */
    UCB0CTLW0 &= ~UCSWRST;       /* Release from reset */
    return TIKU_I2C_OK;
}
```

I2C Arch: Write & Read

Write transaction

```c
int tiku_i2c_arch_write(
    uint8_t addr,
    const uint8_t *buf, uint16_t len)
{
    UCB0I2CSA = addr;
    UCB0CTLW0 |= UCTR | UCTXSTT;

    for (i = 0; i < len; i++) {
        rc = i2c_wait_flag(UCTXIFG0);
        if (rc != TIKU_I2C_OK)
            return rc;
        UCB0TXBUF = buf[i];
    }

    /* Wait for last byte */
    rc = i2c_wait_flag(UCTXIFG0);
    if (rc != TIKU_I2C_OK)
        return rc;

    UCB0CTLW0 |= UCTXSTP;
    return i2c_wait_stop();
}
```

Read transaction (multi-byte)

```c
int tiku_i2c_arch_read(
    uint8_t addr,
    uint8_t *buf, uint16_t len)
{
    UCB0I2CSA = addr;
    UCB0CTLW0 &= ~UCTR;
    UCB0CTLW0 |= UCTXSTT;

    for (i = 0; i < len; i++) {
        /* STOP before last byte */
        if (i == len - 1)
            UCB0CTLW0 |= UCTXSTP;

        rc = i2c_wait_flag(UCRXIFG0);
        if (rc != TIKU_I2C_OK)
            return rc;
        buf[i] = (uint8_t)UCB0RXBUF;
    }

    return i2c_wait_stop();
}
```

Single-Byte Read

For `len==1`, STOP must be issued right after the address phase completes (special handling in the

I2C Arch: Timeout & NACK Detection

Timeout-guarded flag wait with NACK detection

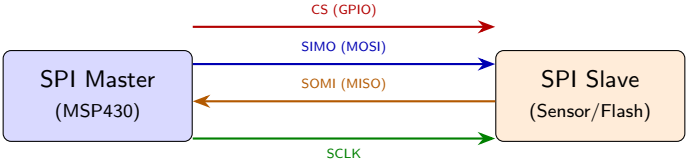
```
static int i2c_wait_flag(uint16_t flag)
{
    uint16_t timeout = 10000U;

    while (!(UCB0IFG & flag)) {
        if (UCB0IFG & UCNACKIFG) {        /* Slave NACKed */
            UCB0IFG &= ~UCNACKIFG;
            UCB0CTLW0 |= UCTXSTP;         /* Issue STOP to release bus */
            while (UCB0CTLW0 & UCTXSTP);
            return TIKU_I2C_ERR_NACK;
        }
        if (--timeout == 0) {
            UCB0CTLW0 |= UCTXSTP;         /* Force STOP on timeout */
            return TIKU_I2C_ERR_TIMEOUT;
        }
    }
    return TIKU_I2C_OK;
}
```

Bus Recovery

Every error path issues a STOP condition to release SDA/SCL, preventing bus lockup. The timeout

SPI Protocol Basics



Four SPI clock modes:

Mode	CPOL	CPHA	Sampling Edge
0	0	0	Rising (most common)
1	0	1	Falling
2	1	0	Falling
3	1	1	Rising

Key characteristics:

- Full-duplex: TX and RX simultaneous
- No addressing — CS selects device
- No ACK/NACK feedback
- Master always generates clock
- Much faster than I2C

SPI Configuration

interfaces/bus/tiku_spi_bus.h

```
/* Clock modes */
#define TIKU_SPI_MODE_0   0
#define TIKU_SPI_MODE_1   1
#define TIKU_SPI_MODE_2   2
#define TIKU_SPI_MODE_3   3

/* Bit order */
#define TIKU_SPI_MSB_FIRST   0
#define TIKU_SPI_LSB_FIRST   1

/* Status codes */
#define TIKU_SPI_OK            0
#define TIKU_SPI_ERR_TIMEOUT  (-1)
#define TIKU_SPI_ERR_PARAM    (-2)

typedef struct tiku_spi_config {
    uint8_t  mode;
    uint8_t  bit_order;
    uint16_t prescaler;
} tiku_spi_config_t;
```

Prescaler convenience macros:
(from `tiku_board_fr5969_launchpad.h`)

Macro	SPI Clock
TIKU_BOARD_SPI_BRW_4MHZ	4 MHz
TIKU_BOARD_SPI_BRW_2MHZ	2 MHz
TIKU_BOARD_SPI_BRW_1MHZ	1 MHz
TIKU_BOARD_SPI_BRW_500KHZ	500 kHz

All derived from 8 MHz SMCLK:
`SPI_CLK = SMCLK / prescaler`

Chip Select

CS is managed by the application via GPIO, not by the SPI driver. This supports arbitrary CS pin assignment and multi-device buses.

SPI API Functions

Function	Description
`tiku_spi_init(config)`	Initialize eUSCI_A1 in 3-pin SPI master mode
`tiku_spi_close()`	Place peripheral in reset
`tiku_spi_transfer(byte)`	Single-byte full-duplex exchange (returns RX byte)
`tiku_spi_write(buf, len)`	Bulk write; received data is discarded
`tiku_spi_read(buf, len)`	Bulk read; sends 0xFF as dummy data
`tiku_spi_write_read(tx, rx, len)`	Bulk full-duplex simultaneous TX and RX

Typical SPI device interaction

```
/* Assert CS, send command, read response, deassert CS */
P3OUT &= ~BIT0;                    /* CS low (active) */
tiku_spi_write(cmd, sizeof(cmd));  /* Send command bytes */
tiku_spi_read(response, 4);        /* Read 4 bytes (sends 0xFF) */
P3OUT |= BIT0;                     /* CS high (inactive) */
```

SPI Board Macros & Mode Mapping

boards/tiku_board_fr5969_launchpad.h

```
/* SPI on eUSCI_A1:
   P2.5=CLK P2.6=SIMO P2.7=SOMI */
#define TIKU_BOARD_SPI_PINS_INIT() \
    do { \
        P2SEL1 |= BIT5|BIT6|BIT7; \
        P2SEL0 &= ~(BIT5|BIT6|BIT7);\
    } while(0)

/* Prescalers: 8 MHz SMCLK */
#define TIKU_BOARD_SPI_BRW_4MHZ   2
#define TIKU_BOARD_SPI_BRW_2MHZ   4
#define TIKU_BOARD_SPI_BRW_1MHZ   8
#define TIKU_BOARD_SPI_BRW_500KHZ 16
```

MSP430 SPI mode mapping:

MSP430's `UCCKPH` is **inverted** relative to the standard CPHA definition.

SPI Mode	UCCKPL	UCCKPH	Bits
Mode 0	0	1	`UCCKPH`
Mode 1	0	0	0
Mode 2	1	1	`UCCKPH|UCCKPL`
Mode 3	1	0	`UCCKPL`

Common Pitfall

Standard CPHA=0 maps to MSP430 UCCKPH=1. The driver handles this mapping internally — just use `TIKU_SPI_MODE_0..3`.

SPI Arch: Initialization

arch/msp430/tiku_spi_arch.c — tiku_spi_arch_init()

```c
int tiku_spi_arch_init(const tiku_spi_config_t *config)
{
    /* Select SPI function on board-specific SCLK/SIMO/SOMI pins */
    TIKU_BOARD_SPI_PINS_INIT();

    /* Put eUSCI_A1 in reset before configuration */
    UCA1CTLW0 = UCSWRST;

    /* SPI master, synchronous, 3-pin mode, SMCLK source */
    UCA1CTLW0 |= UCMST | UCSYNC | UCMODE_0 | UCSSEL__SMCLK;

    /* Set clock polarity and phase for requested SPI mode */
    UCA1CTLW0 |= spi_mode_to_bits(config->mode);

    /* Set bit order */
    if (config->bit_order == TIKU_SPI_MSB_FIRST)
        UCA1CTLW0 |= UCMSB;

    /* Set clock prescaler */
    UCA1BRW = config->prescaler;

    UCA1CTLW0 &= ~UCSWRST;  /* Release from reset */
    return TIKU_SPI_OK;
}
```

SPI Arch: Transfer Operations

Single-byte full-duplex exchange

```c
static uint8_t spi_xfer_byte(uint8_t tx)
{
    uint16_t timeout;

    timeout = SPI_TIMEOUT;
    while (!(UCA1IFG & UCTXIFG)) {
        if (--timeout == 0)
            return 0;
    }
    UCA1TXBUF = tx;

    timeout = SPI_TIMEOUT;
    while (!(UCA1IFG & UCRXIFG)) {
        if (--timeout == 0)
            return 0;
    }
    return (uint8_t)UCA1RXBUF;
}
```

Bulk read (sends 0xFF dummy)

```c
int tiku_spi_arch_read(
    uint8_t *buf, uint16_t len)
{
    uint16_t i, timeout;

    for (i = 0; i < len; i++) {
        timeout = SPI_TIMEOUT;
        while (!(UCA1IFG & UCTXIFG))
            if (--timeout == 0)
                return TIKU_SPI_ERR_TIMEOUT;
        UCA1TXBUF = 0xFF;

        timeout = SPI_TIMEOUT;
        while (!(UCA1IFG & UCRXIFG))
            if (--timeout == 0)
                return TIKU_SPI_ERR_TIMEOUT;
        buf[i] = (uint8_t)UCA1RXBUF;
    }
    return TIKU_SPI_OK;
}
```

SPI is Master-Clocked

Unlike I2C, SPI has no bus contention — the master always generates the clock. Timeouts guard against hardware faults only.

Example: Reading a Temperature Sensor (I2C)

I2C sensor read — TMP102 temperature sensor at address 0x48

```c
#define TMP102_ADDR     0x48
#define TMP102_REG_TEMP 0x00

static void read_temperature(void)
{
    tiku_i2c_config_t cfg = { .speed = TIKU_I2C_SPEED_STANDARD };
    uint8_t reg = TMP102_REG_TEMP;
    uint8_t buf[2];
    int rc;

    tiku_i2c_init(&cfg);

    /* Write register address, then read 2 bytes (repeated START) */
    rc = tiku_i2c_write_read(TMP102_ADDR, &reg, 1, buf, 2);

    if (rc == TIKU_I2C_OK) {
        /* TMP102: 12-bit result, MSB first, 0.0625 deg C per LSB */
        int16_t raw = ((int16_t)buf[0] << 4) | (buf[1] >> 4);
        TIKU_PRINTF("Temp: %d.%02d C\n", raw / 16, (raw % 16) * 625 / 100);
    } else {
        TIKU_PRINTF("I2C error: %d\n", rc);
    }

    tiku_i2c_close();
}
```

Example: SPI Flash Read

SPI flash read — W25Q SPI flash memory

```
#define FLASH_CS_LOW()   do { P3OUT &= ~BIT0; } while(0)
#define FLASH_CS_HIGH()  do { P3OUT |=  BIT0; } while(0)
#define FLASH_CMD_READ   0x03

static void flash_read(uint32_t addr, uint8_t *buf, uint16_t len)
{
    tiku_spi_config_t cfg = {
        .mode      = TIKU_SPI_MODE_0,
        .bit_order = TIKU_SPI_MSB_FIRST,
        .prescaler = TIKU_BOARD_SPI_BRW_2MHZ
    };
    uint8_t cmd[4];

    tiku_spi_init(&cfg);

    /* Build 4-byte read command: opcode + 24-bit address */
    cmd[0] = FLASH_CMD_READ;
    cmd[1] = (uint8_t)(addr >> 16);
    cmd[2] = (uint8_t)(addr >> 8);
    cmd[3] = (uint8_t)(addr);

    FLASH_CS_LOW();
    tiku_spi_write(cmd, 4);       /* Send command + address */
    tiku_spi_read(buf, len);      /* Clock out data (sends 0xFF) */
```

Example: Process-Based Sensor Polling

Periodic I2C sensor read using TikuOS process + timer

```c
#define SENSOR_ADDR 0x48
#define SENSOR_REG  0x00

static struct tiku_timer poll_timer;

TIKU_PROCESS(sensor_process, "I2C Sensor");

TIKU_PROCESS_THREAD(sensor_process, ev, data)
{
    static tiku_i2c_config_t cfg = { .speed = TIKU_I2C_SPEED_STANDARD };
    static uint8_t reg = SENSOR_REG;
    static uint8_t buf[2];

    TIKU_PROCESS_BEGIN();

    tiku_i2c_init(&cfg);
    tiku_timer_set_event(&poll_timer, TIKU_CLOCK_SECOND * 2);

    while (1) {
        TIKU_PROCESS_WAIT_EVENT_UNTIL(ev == TIKU_EVENT_TIMER);

        if (tiku_i2c_write_read(SENSOR_ADDR, &reg, 1, buf, 2) == TIKU_I2C_OK)
            TIKU_PRINTF("Sensor: 0x%x%x\n", buf[0], buf[1]);

        tiku_timer_reset(&poll_timer);  /* drift-free 2s interval */
    }

    TIKU_PROCESS_END();
}
TIKU_AUTOSTART_PROCESSES(&sensor_process);
```

Bus Interface Module Files

Layer	File	Role
I2C API	`interfaces/bus/tiku_i2c_bus.h` `interfaces/bus/tiku_i2c_bus.c`	Public types & prototypes Validation, delegates to arch
SPI API	`interfaces/bus/tiku_spi_bus.h` `interfaces/bus/tiku_spi_bus.c`	Public types & prototypes Validation, delegates to arch
HAL	`hal/tiku_i2c_hal.h` `hal/tiku_spi_hal.h`	Routes to I2C arch Routes to SPI arch
Arch (MSP430)	`arch/msp430/tiku_i2c_arch.c/.h` `arch/msp430/tiku_spi_arch.c/.h`	eUSCI_B0 I2C master eUSCI_A1 SPI master
Board Config	`.../tiku_device_fr5969.h` `.../tiku_board_fr5969_launchpad.h`	eUSCI availability flags Pin select & prescaler macros

Summary

I2C Bus

- 2-wire (SDA + SCL) via eUSCI_B0
- 100 kHz standard / 400 kHz fast
- 3 operations: write, read, write_read
- NACK detection + timeout guards
- Auto STOP on every error path

SPI Bus

- 3-wire + CS via eUSCI_A1
- All 4 SPI modes, MSB/LSB order
- 4 operations: transfer, write, read, write_read
- Application-managed CS via GPIO
- Up to 4 MHz from 8 MHz SMCLK

Architecture

- Platform-independent API in `interfaces/bus/`
- HAL routing in `hal/`
- MSP430 arch in `arch/msp430/`
- Board macros for pin config & clock
- Same arch code works across all FR variants

FR5969 LaunchPad Module Assignment

eUSCI_A0	UART (backchannel)
eUSCI_A1	SPI
eUSCI_B0	I2C
eUSCI_B1	(available)

Thank You

TikuOS Bus Interfaces

I2C & SPI for Physical AI Endpoints

Repository:	`https://github.com/tikuos/tikuOS`
License:	Apache 2.0
I2C API:	`interfaces/bus/tiku_i2c_bus.h`
SPI API:	`interfaces/bus/tiku_spi_bus.h`
I2C Arch:	`arch/msp430/tiku_i2c_arch.c`
SPI Arch:	`arch/msp430/tiku_spi_arch.c`